

Final Exam

Quantitative Methods for Business

Daniel Sánchez Pazmiño

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Instructions

This exam lasts one hour and a half (90 minutes). Please answer in the Google form [here](#). It is mandatory to keep your camera on during the entire exam. You are not allowed to use any external resources.

Questions

1. A new statistical software has been developed. The following operation was performed in the console:

$$2 + 5 + 6 \div 3 \times 2$$

The answer to the operation provided by the software was **8.67**. What order of operations seems to be used by the software?

- a. PEMDAS: Parentheses, Exponents, Multiplication and Division (from left to right), Addition and Subtraction (from left to right).
- b. Addition, Subtraction, Multiplication, Division.
- c. Division, Multiplication, Addition, Subtraction.
- d. None of the above.

2. A study involved implementing different treatments to different towns around a certain province in country. The treatment involved the use of political awareness campaigns to see if the number of people voting in the next election increased. 50% of people registered to vote was randomly selected to receive the treatment. Within the treatment group, 60% of people lived in urban areas. How much of those registered to vote received the treatment and lived in urban areas?

- a. 30%
- b. 20%
- c. 12%
- d. None of the above

3. When one increases a value by a certain percentage, and then decreases the result by the same percentage, the final value is always less than the original value.

- a. True
- b. False
- c. Cannot be determined
- d. None of the above

4. The domain of the function $f(x) = \sqrt{4 - x^2}$ is:

- a. $x \in \mathbb{R}$ (all real numbers)
- b. $x \in \mathbb{R} \mid x \leq 2$ (all real numbers less than or equal to 2)
- c. $x \in \mathbb{R} \mid -2 \leq x \leq 2$ (all real numbers between -2 and 2)
- d. None of the above

5. For the equation $AK\sqrt{L} = Y$, where A , K , Y are parameters and L is a variable, the solution for L is:

- a. $L = \frac{Y}{AK}$
- b. $L = \frac{Y^2}{AK}$
- c. $L = \frac{Y^2}{AK^2}$
- d. None of the above

6. The solution to $x^2 - 80 = 1$ is

- a. $x = 9$

- b. $x = 10$
- c. $x = 11$
- d. None of the above

7. The population variance formula is given by:

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

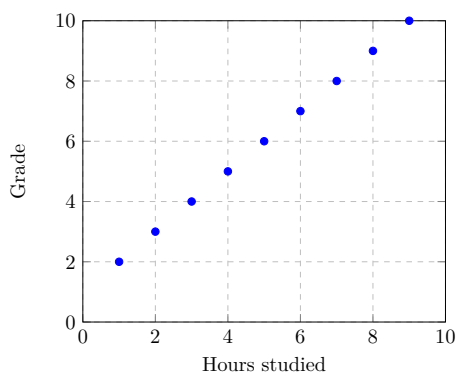
Where $\bar{x}$ is the sample mean. For calculating the sample variance, a “Bessel’s correction” is applied, where the denominator is $n - 1$ instead of n . This is equal to:

- a. $\sum_{i=1}^n \frac{1}{n-1} (x_i - \bar{x})^2$
- b. $\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$
- c. Both a and b
- d. None of the above

8. The expression $(2^3 \times 2^4 \div 8) \times (16^{1/2} \div 2)$ is equal to

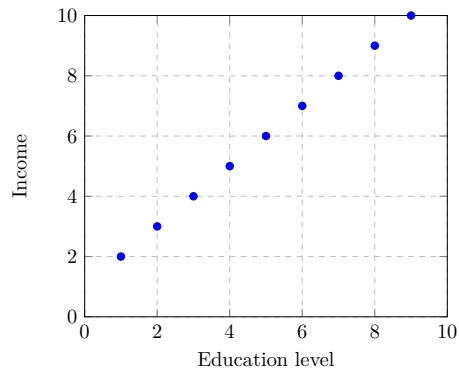
- a. 16
- b. 32
- c. 2^3
- d. 2^2

9. The following is a scatter plot of the relationship between the number of hours studied and the grade obtained in a test. What can be said about the relationship between the two variables?



- a. The variables are positively related to each other and they are in the first quadrant.

- b. The variables are negatively related to each other and they are in the first quadrant.
 - c. The variables are positively related to each other and they are in the second quadrant.
 - d. None of the above.
10. From the above graph, what is the estimated line which passes through the points?
- a. $y = 2x$
 - b. $y = x$
 - c. $y = x + 1$
 - d. $y = x + 1$
11. The vertex of the parabola $y = x^2 - 4x + 3$ is:
- a. $(2, 1)$
 - b. $(2, -1)$
 - c. $(2, 3)$
 - d. $(2, -1)$
12. The graph above shows some data about education level and income. Given the observed relationship, what scale on the y-axis would imply that income is growing at a 10% constant rate for every unit increase in education level?



- a. Regular scale
- b. Natural log scale
- c. Applying a 10% increase to the y-axis
- d. None of the above

13. The solution to the equation above for t is:

$$\frac{1}{\sqrt{2\pi}}e^{-1/2t^2} = \frac{1}{8}$$

a. $t = \sqrt{\ln \frac{32}{\pi}}$

b. $t = \sqrt{\ln \frac{16}{\pi}}$

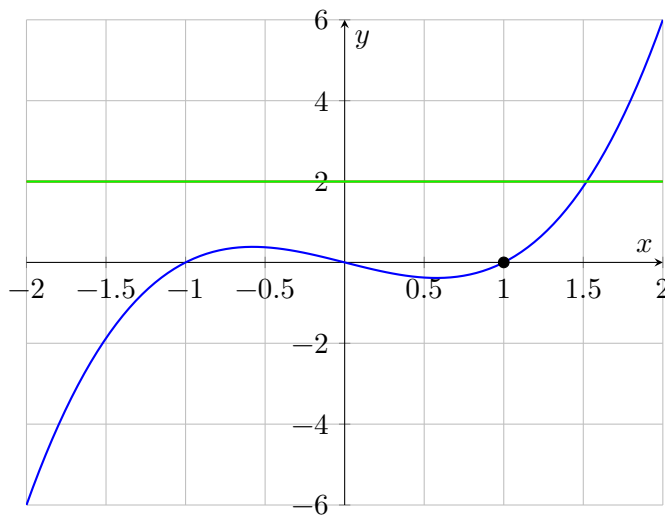
c. $t = -\sqrt{\ln \frac{32}{\pi}}$

d. None of the above

14. A political scientist has theorized that the popularity function of a candidate will be given determined in a complex model, where popularity is a function of the number of votes obtained in the previous election, the number of years in office, and the number of scandals in which the candidate has been involved. A reduced form estimation would potentially involve:

- a. Estimating the popularity function directly, using a simplified model.
- b. Carefully determining the relationships between the variables in separate models, taking the theoretical model seriously.
- c. No estimation at all, as the model is too complex to be estimated.
- d. None of the above.

15. Consider the graph below.



The line at $y = 2$ is tangent to the curve $y = x^3 - x$.

- a. True

- b. False
 - c. Cannot be determined
 - d. None of the above
16. The slope of the tangent line to the curve $y = x^3 - x$ at $x = 1$ is:
- a. 2
 - b. 3
 - c. 4
 - d. 2
17. The elasticity of jail time with respect to police presence (in number of officers) was estimated as -0.05. The interpretation of this coefficient is:
- a. For every additional police officer, jail time decreases by 0.05 units.
 - b. For every additional police officer, jail time increases by 0.05 units.
 - c. For every additional police officer, jail time increases by 5%.
 - d. For every additional police officer, jail time decreases by 5%.
18. It was observed that police presence would increase in a certain town from 100 to 200 officers. The expected change in jail time would be:
- a. -5%
 - b. 5%
 - c. -10%
 - d. None of the above
19. The satisfaction from a criminal from committing a crime was estimated by the function $U = 10 + 2C^2$, where C is the number of years in jail. At what level of jail time is satisfaction minimized?
- a. $C = 0$
 - b. $C = 1$
 - c. $C = 2$
 - d. None of the above
20. The gradient of function $f(x, y) = x^2 + \ln(y^2)$ at the point $(1, 1)$ is:
- a. $(2, 2)$
 - b. $(2, 0)$
 - c. $(2, 0.5)$
 - d. None of the above